MATHEMATICS FOR EVERYONE

For Mathematics
and
For all my fellow Mathematicians

Contents

0 The World of Mathematics **1**

1 Reading Mathematics **3**

2 Drawing Mathematics **5**

3 Writing Mathematics **7**

4 Logic **9**

5 Sets **11**

6 Relations **13**

7 Functions **15**

Chapter 0

The World of Mathematics

Chapter 1

Reading Mathematics

Chapter 2

Drawing Mathematics

Chapter 3

Writing Mathematics

Chapter 4

Logic

Chapter 5

Sets

Definition: Even Integer

An integer n is called even if there exists an integer k such that $n = 2k$.

Exercises

1. Prove that $A = B$, where

$$A = \{m \in \mathbb{Z} \mid m = 2a \text{ for some integer } a\}$$

and

$$B = \{n \in \mathbb{Z} \mid n = 2b - 2 \text{ for some integer } b\}.$$

Chapter 6

Relations

Chapter 7

Functions